

HARDIK AGARWAL

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Education

Vellore Institute of Technology, Bhopal

B.Tech Computer Science Engineering, CGPA: 8.53

2023 – Present

Bhopal, India

St. Joseph's College, Prayagraj

Class XII: 91.25%

2023

St. Joseph's College, Prayagraj

ICSE Class X: 93.6%

2021

Technical Skills

Languages: Python, Java, SQL, JavaScript

Machine Learning: TensorFlow, Keras, Scikit-learn, CNN, LSTM, GRU, Deep Learning

AI: Retrieval-Augmented Generation (RAG), LLMs, Whisper, Ollama, BGE-M3, Semantic Search

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Time Series Forecasting

Research: Causal Inference, Convergent Cross Mapping (CCM), Model Evaluation

Web/Tools: Next.js, React, Flask, Django, REST APIs, AWS, Git, GitHub

Work Experience

Motilal Nehru National Institute of Technology Allahabad

25 May 2026 – 21 Jun 2026

Research Intern (Supervisor: Prof. Anil Kumar Singh)

Prayagraj, India

- Reproduced and extended a causality-powered stock forecasting framework using Pearson Correlation and Convergent Cross Mapping (CCM) on Indian equity markets.
- Built an end-to-end preprocessing pipeline covering chronological splitting, normalization and sliding-window feature generation for two years of financial time-series data.
- Designed and evaluated DNN, CNN, LSTM and GRU models using TensorFlow/Keras with RMSE, MAE, MAPE and R^2 metrics.
- Demonstrated superior forecasting using causal closing-price features over full OHLCV inputs through extensive experimentation and technical reporting.

Projects

RAG-Based Educational Video Assistant

Python — Whisper — Ollama

- Built a Retrieval-Augmented Generation system enabling semantic search across course videos with timestamp-aware recommendations.
- Implemented Whisper Large-v2 transcription, BGE-M3 embeddings, cosine similarity retrieval and Llama 3.2 local inference via Ollama.
- Automated transcript generation, embedding creation and context-aware responses without external APIs.

Smart-Seva — Frontend — Backend — Live

Next.js — Flask

- Developed a full-stack agricultural decision support platform integrating weather, crop pricing and educational APIs.
- Built analytics pipelines, ML inference modules and secure RBAC-based authentication with modular frontend/backend architecture.
- Designed for independent deployment and scalable maintenance using separate repositories.

Certifications & Recognition

- Research Internship Completion Certificate – MNNIT Allahabad
- NPTEL Elite Certification – Machine Learning (IIT Madras)
- Google – Bits and Bytes of Computer Networking
- HackerRank Gold Badges – Java, Python and C++
- Graphs Camp Participation Certificate – AlgoUniversity (Oct 2025)

Achievements

- Solved 120+ Data Structures and Algorithms problems across coding platforms.
- Research interests include Deep Learning, Financial AI, Generative AI and Time Series Forecasting.